

Proposition 24: For any $\gamma > 0$,

$$\sup_{\hat{f}: \Pi \rightarrow \mathbb{R}} \text{dec}_{\gamma}(\mathcal{F}, \hat{f}) = \sup_{\hat{f} \in \text{co}(\mathcal{F})} \text{dec}_{\gamma}(\mathcal{F}, \hat{f}).$$

4.7 Exercises

Exercise 8 (Posterior Sampling for Multi-Armed Bandits): Prove that for the standard multi-armed bandit,

$$\underline{\text{dec}}_{\gamma}(\mathcal{F}) \lesssim \frac{|\Pi|}{\gamma},$$

by using the Posterior Sampling strategy (select p to be the action distribution induced by sampling $f \sim \mu$ and selecting π_f), and applying the decoupling lemma (Lemma 9). Recall that here, $\underline{\text{dec}}_{\gamma}(\mathcal{F})$ is the “maxmin” version of the DEC (4.28).

Exercise 9: Prove Proposition 19.

Exercise 10: In this exercise, we will prove Proposition 24 as follows. First, show that the left-hand side is an upper bound on the right-hand side. For the other direction:

1. Prove that

$$\inf_{\hat{f} \in \text{co}(\mathcal{F})} \mathbb{E}_{f \sim \mu} \mathbb{E}_{\pi \sim p} (f(\pi) - \hat{f}(\pi))^2 \leq \inf_{\hat{f}: \Pi \rightarrow \mathbb{R}} \mathbb{E}_{f \sim \mu} \mathbb{E}_{\pi \sim p} (f(\pi) - \hat{f}(\pi))^2. \quad (4.31)$$

2. Use the Minimax Theorem (Lemma 41 in Appendix A.3) to conclude Proposition 24.

5. REINFORCEMENT LEARNING: BASICS

We now introduce the framework of *reinforcement learning*, which encompasses a rich set of dynamic, stateful decision making problems. Consider the task of repeated medical treatment assignment, depicted in Figure 4. To make the setting more realistic, it is natural to allow the decision-maker to apply *multi-stage strategies* rather than simple one-shot decisions such as “prescribe a painkiller.” In principle, in the language of structured bandits, nothing is preventing us from having each decision π^t be a *complex multi-stage treatment strategy* that, at each stage, acts on the patient’s *dynamic state*, which evolves as a *function of the treatments at previous stages*. As an example, intermediate actions of the type “if patient’s blood pressure is above X then do Y” can form a decision tree that defines the complex strategy π^t . Methods from the previous lectures provide guarantees for such a setting, as long as we have a succinct model of expected rewards. What sets RL apart from structured bandits is the *additional information* about the *intermediate state transitions* and *intermediate rewards*. This information facilitates *credit assignment*, the mechanism for recognizing *which* of the *actions* led to the *overall* (composite) *decision* to be good or bad. This extra information can reduce what would otherwise be exponential sample complexity in terms of the number of stages, states, and actions in *multi-stage decision making*.

< Credit Assignment >

내려온 수많은 decision(action) 중
“정제적인” 결과를 달성 높이 수 있게 recognize
} → Reduce Complexity
(in # of stages, states, actions)

이전 stage에서의
treatment에서 기반한
현재의 환자의 state
(dynamic state)에 따라
새로운 decision을 할함.

This section is structured as follows. We first present the formal reinforcement learning framework and present basic principles including Bellman optimality and dynamic programming, which facilitate efficiently computing optimal decisions when the environment is known. We then consider the case in which the environment is unknown, and give algorithms for perhaps the simplest reinforcement learning setting, tabular reinforcement learning, where the state and action spaces are finite (and small). Algorithms for more complex reinforcement learning settings are given in Section 5.

환경이 알려져 있을 때
optimal decision을 효율적으로 compute

① RL framework, Bellman optimality, DP

②: 환경이 알려지지 않은 경우

③: "tabular RL" 시스템에서의 알고리즘
↳ state & action spaces finite

5.1 Finite-Horizon Episodic MDP Formulation

We consider an episodic finite-horizon reinforcement learning framework. With H denoting the *horizon*, a *Markov Decision Process* (MDP) M takes the form

<MDP>
Each event depends SOLELY on the state attained in the previous event.

$$M = \{ \mathcal{S}, \mathcal{A}, \{P_h^M\}_{h=1}^H, \{R_h^M\}_{h=1}^H, d_1 \},$$

$\mathcal{S}$: State Space $\mathcal{A}$: Action Space P_h^M : Probability transition kernel R_h^M : Reward Distribution d_1 : Initial state dist.
 (vary across MDP) (fixed and known)

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space,

$$P_h^M : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$$

is the probability transition kernel at step h ,

$$R_h^M : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathbb{R})$$

is the reward distribution, and $d_1 \in \Delta(\mathcal{S})$ is the initial state distribution. We allow the reward distribution and transition kernel to vary across MDPs, but assume for simplicity that the initial state distribution is fixed and known.

For a fixed MDP M , an episode proceeds under the following protocol. At the beginning of the episode, the learner selects a randomized, non-stationary policy

$$\pi = (\pi_1, \dots, \pi_H),$$

where $\pi_h : \mathcal{S} \rightarrow \Delta(\mathcal{A})$; (we let Π_{rns} for "randomized, non-stationary" denote the set of all such policies.) The episode then evolves through the following process, beginning from

② State 1 $s_1 \sim d_1$. For $h = 1, \dots, H$: (In one episode)
(initial state dist) Horizon

- $a_h \sim \pi_h(s_h)$.
action Policy(state)
- $r_h \sim R_h^M(s_h, a_h)$ and $s_{h+1} \sim P_h^M(s_h, a_h)$.
reward State action Prob. transition kernel

For notational convenience, we take s_{H+1} to be a deterministic terminal state. The Markov property refers to the fact that under this evolution,

★ Memoryless property.

$$\mathbb{P}^M(s_{h+1} = s' | s_h, a_h) = \mathbb{P}^M(s_{h+1} = s' | s_h, a_h, s_{h-1}, a_{h-1}, \dots, s_1, a_1).$$

↳ solely depend on the state at time h // independent of the state of the process at any time PREVIOUS to h .

★ The value for a policy π under M is given by

$$f^M(\pi) := \mathbb{E}^{M, \pi} \left[\sum_{h=1}^H r_h \right], \quad (5.1)$$

Value for policy π reward

where $\mathbb{E}^{M, \pi}[\cdot]$ denotes expectation under the process above. We define an optimal policy for model M as

★

$$\pi_M \in \arg \max_{\pi \in \Pi_{\text{rns}}} f^M(\pi). \quad (5.2)$$

Optimal policy for model M A particular MDP may have multiple distinct optimal policies

Optimal Policy
 $\pi_M \in \arg \max_{\pi \in \Pi_M} f^M(\pi)$

Value functions. Maximization in (5.2) is a daunting task, since each policy π is a complex multi-stage object. It is useful to define intermediate “reward-to-go” functions to start breaking this complex task into smaller sub-tasks. Specifically, for a given model M and policy π , we define the **state-action value function** and **state value function** via

★ (State)action value function State value function

$$Q_h^{M,\pi}(s, a) = \mathbb{E}^{M,\pi} \left[\sum_{h'=h}^H r_{h'} \mid s_h = s, a_h = a \right], \quad \text{and} \quad V_h^{M,\pi}(s) = \mathbb{E}^{M,\pi} \left[\sum_{h'=h}^H r_{h'} \mid s_h = s \right]$$

행동 가치 함수 상태 가치 함수

Hence, the definition in (5.1) reads

$$f^M(\pi) := \mathbb{E}^{M,\pi} \left[\sum_{h=1}^H r_h \right] \quad f^M(\pi) = \mathbb{E}_{s \sim d_1, a \sim \pi_1(s)} [Q_1^{M,\pi}(s, a)] = \mathbb{E}_{s \sim d_1} [V_1^{M,\pi}(s)]$$

Value for policy π

★ $f^M(\pi) = \mathbb{E}^{M,\pi} \left[\sum_{h=1}^H r_h \right]$

State value function

★ $V_h^{M,\pi}(s) = \mathbb{E}^{M,\pi} \left[\sum_{h'=h}^H r_{h'} \mid s_h = s \right]$

State-action value function

★ $Q_h^{M,\pi}(s, a) = \mathbb{E}^{M,\pi} \left[\sum_{h'=h}^H r_{h'} \mid s_h = s, a_h = a \right]$

Handwritten notes:

- State value function: $V_h(s) = \mathbb{E}_\pi [G_t | s_t = s]$ (정리)
- $= \mathbb{E}_\pi [r_t + \gamma V_{h+1}(s_{t+1}) | s_t = s]$
- $= \mathbb{E}_\pi [r_t + \gamma (r_{t+1} + \gamma V_{h+2}(s_{t+2})) | s_t = s]$
- $= \mathbb{E}_\pi [r_t + \gamma (G_{t+1}) | s_t = s]$
- $= \mathbb{E}_\pi [r_t + \gamma V(s_{t+1}) | s_t = s]$

Online RL. For reinforcement learning, our main focus will be on what is called the **online reinforcement learning problem**, in which we interact with an **unknown MDP** M^* for T episodes. For each episode $t = 1, \dots, T$, the learner selects a policy $\pi^t \in \Pi_{\text{rms}}$. The policy is executed in the MDP M^* , and the learner observes the resulting trajectory

resulting trajectory (on episode t)

$$\tau^t = (s_1^t, a_1^t, r_1^t), \dots, (s_H^t, a_H^t, r_H^t).$$

★ The goal is to minimize the total regret

Optimal policy policy on episode t

$$\sum_{t=1}^T \mathbb{E}_{\pi^t \sim p^t} [f^{M^*}(\pi^{M^*}) - f^{M^*}(\pi^t)] \quad (5.3)$$

against the optimal policy π^{M^*} for M^* .

The online RL framework is a strict generalization of (structured) **bandits** and **contextual bandits** (with i.i.d. contexts). Indeed, if $\mathcal{S} = \{s_0\}$ and $H = 1$, each episode amounts to choosing an action $a^t \in \mathcal{A}$ and observing a reward r^t with mean $f^M(a^t)$, which is precisely a **bandit** problem. On the other hand, taking $\mathcal{S} = \mathcal{X}$ and $H = 1$ puts us in the setting of **contextual bandits**, with d_1 being the distribution of contexts. In both cases, the notion of regret (5.3) coincides with the notion of regret in the respective setting.

We mention in passing that many alternative formulations for Markov decision processes and for the reinforcement learning problem appear throughout the literature. For example, MDPs can be studied with infinite horizon (with or without discounting), and an alternative to minimizing regret is to consider **PAC-RL** which aims to minimize the sub-optimality of a final output policy produced after exploring for T rounds.

5.2 Planning via Dynamic Programming

In some reinforcement learning problems, it is natural to assume that the **true MDP** M^* is **known**. This may be the case with games, such as **chess** or backgammon, where **transition probabilities** are postulated by the game itself. In other settings, such as robotics or medical treatment, the agent interacts with an **unknown** M^* and needs to learn at least some aspects of this environment. The online reinforcement learning problem described above falls in the latter category. Before attacking the learning problem, we need to understand the structure of solutions to (5.2) in the case where M^* is known to the decision-maker. In this section, we show that the problem of maximizing $f^M(\pi)$ over $\pi \in \Pi$ in a **known model** M (known as **planning**) can be solved efficiently via the principle of **dynamic programming**. Dynamic

transition probabilities are known... M^* known

Online RL: True MDP M^* is unknown!!

Optimal policy 찾기

1회: 1회 (Horizon)
 각 episode (회차) 별로, 선택하는 policy를 다르게 함.
 그렇게 여러 회차 동안 다 돌리면서 (1회)
 시뮬레이션하고,
 총 T개의 결과를 사용해서 평가. 즉, 총 T회차.

Ch2: Planning via DP
maximizing $f_v(\pi)$ (value) over $\pi \in \Pi$
in a MDP model M

Dynamic Programming:
복잡한 multi-stage decision을
여러 개의 작은 decision들의 배열...로 쪼개면서
credit assignment 문제를 해결하는 것.

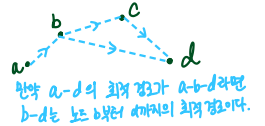
programming can be viewed as solving the problem of credit assignment by breaking down a complex multi-stage decision (policy) into a sequence of small decisions.

We start by observing that the optimal policy π_M in (5.2) may not be uniquely defined. For instance, if d_1 assigns zero probability to some state s_1 , the behavior of π_M on this state is immaterial.

In what follows, we introduce a fundamental result, Proposition 25, which guarantees existence of an optimal policy $\pi_M = (\pi_{M,1}, \dots, \pi_{M,H})$ that maximizes $V_1^{M,\pi}(s)$ over $\pi \in \Pi_{\text{rns}}$ for all states $s \in \mathcal{S}$ simultaneously (rather than just on average, as in (5.2)).

The fact that such a policy exists may seem magical at first, but it is rather straightforward. Indeed, if $\pi_{M,h}(s)$ is defined for all $s \in \mathcal{S}$ and $h = 2, \dots, H$, then defining the optimal $\pi_{M,1}(s)$ at any s is a matter of greedily choosing an action that maximizes the sum of the expected immediate reward and the remaining expected reward under the optimal policy. Indeed, this observation is Bellman's principle of optimality, stated more generally as follows [17]:

PRINCIPLE OF OPTIMALITY. An optimal policy has the property that whatever the initial state and initial decisions are, the remaining decisions must constitute an optimal policy (with regard to the state resulting from the first decisions.)



말이 다
최적의 정책은 초기의 상태와 결정이 무엇이었는지와 관계없이 남아있는 결정들이 초기의 결정들에 의해
교수를 기망하여 출석을 안한 것은 죄가 아니지만 나쁘다고 할 수

최적의 정책이라 함은!
- 어떤 (최적의) state와 decision이 주어진다면 상관없이 resulting from... the first decision
남아 있는 (그 이후의) 결정들은, 이런 결정으로 인해 초래된 상태에 대해 최적의 결정이다.

To state the result formally, we introduce the optimal value functions as

Optimal Value Functions
(of each s, a, h)
↓
이 "가장"을
"동시에" 최대화하는
single policy가 존재하냐?

$$Q_h^{M,*}(s, a) = \max_{\pi \in \Pi_{\text{rns}}} \mathbb{E}^{M,\pi} \left[\sum_{h'=h}^H r_{h'} \mid s_h = s, a_h = a \right] \quad \text{and} \quad V_h^{M,*}(s) = \max_{a \in \mathcal{A}} Q_h^{M,*}(s, a) \quad (5.4)$$

for all $s \in \mathcal{S}$, $a \in \mathcal{A}$, and $h \in [H]$; we adopt the convention that $V_{H+1}^{M,*}(s) = Q_{H+1}^{M,*}(s, a) = 0$. Since these optimal values are separate maximizations for each s, a, h , it is reasonable to ask whether there exists a single policy that maximizes all these value functions simultaneously. Indeed, the following lemma shows that there exists π_M such that for all s, a, h ,

$$Q_h^{M,*}(s, a) = Q_h^{M,\pi_M}(s, a), \quad \text{and} \quad V_h^{M,*}(s) = V_h^{M,\pi_M}(s). \quad \forall s, a, h \quad (5.5)$$

Proposition 25 (Bellman Optimality): The optimal value function (5.4) for MDP M can be computed via $V_{H+1}^{M,\pi_M}(s) := 0$, and for each $s \in \mathcal{S}$,

$$V_h^{M,\pi_M}(s) = \max_{a \in \mathcal{A}} \mathbb{E}^M [r_h + V_{h+1}^{M,\pi_M}(s_{h+1}) \mid s_h = s, a_h = a]. \quad (5.6)$$

즉, 현 시점의 state 가치함수가 바로 다음 시점의 state 가치함수로 표현된다!

The optimal policy is given by

$$\pi_{M,h}(s) \in \arg \max_{a \in \mathcal{A}} \mathbb{E}^M [r_h + V_{h+1}^{M,\pi_M}(s_{h+1}) \mid s_h = s, a_h = a]. \quad (5.7)$$

Equivalently, for all $s \in \mathcal{S}$, $a \in \mathcal{A}$,

$$Q_h^{M,\pi_M}(s, a) = \mathbb{E}^M \left[r_h + \max_{a' \in \mathcal{A}} Q_{h+1}^{M,\pi_M}(s_{h+1}, a') \mid s_h = s, a_h = a \right], \quad (5.8)$$

(compute the maximum over ALL actions!)

and an the **optimal policy** is given by

$$\pi_{M,h}(s) \in \arg \max_{a \in \mathcal{A}} Q_h^{M, \pi_M}(s, a). \quad (5.9)$$

$$Q_h^{M, \pi_M}(s, a) = \mathbb{E}^M [r_h + \max_{a' \in \mathcal{A}} Q_{h+1}^{M, \pi_M}(s_{h+1}, a') \mid s_h = s, a_h = a]$$

The update in (5.8) is referred to as **value iteration** (VI). It is useful to introduce a more succinct notation for this update. For an MDP M , define the **Bellman Operators** $\mathcal{T}_1^M, \dots, \mathcal{T}_H^M$ via

$$[\mathcal{T}_h^M Q](s, a) = \mathbb{E}_{s_{h+1} \sim P_h^M(s, a), r_h \sim R_h^M(s, a)} \left[r_h(s, a) + \max_{a' \in \mathcal{A}} Q(s_{h+1}, a') \right] \quad (5.10)$$

for any $Q : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. Going forward, we will write the expectation above more succinctly as

$$[\mathcal{T}_h^M Q](s, a) = \mathbb{E}^M \left[r_h(s_h, a_h) + \max_{a' \in \mathcal{A}} Q(s_{h+1}, a') \mid s_h = s, a_h = a \right] \quad (5.11)$$

In the language of Bellman operators, (5.8) can be written as

$$Q_h^{M, \pi_M} = \underbrace{\mathcal{T}_h^M}_{\text{Bellman Operators}} Q_{h+1}^{M, \pi_M}. \quad (5.12)$$

5.3 Failure of Uniform Exploration

The task of planning using **dynamic programming**—which requires knowledge of the MDP—is fairly straightforward, at least if we **disregard** the **computational concerns**. In this course, however, we are interested in the problem of learning to make decisions in the face of an **unknown environment**. Minimizing regret (in an unknown MDP) requires **“exploration.”** As the next example shows, **exploration** in MDPs is a more delicate issue than in bandits.

Recall that ε -Greedy, a simple algorithm, is a reasonable solution for bandits and contextual bandits, albeit with a suboptimal rate ($T^{2/3}$ as opposed to $\sqrt{T}$). The next (classical) example, a so-called **“combination lock,”** shows that such a strategy can be disastrous in **reinforcement learning**, as it leads to exponential (in the horizon H) regret.

Consider the MDP depicted in **Figure 8**, with $H + 2$ states, and two actions a_g and a_b , and a starting state 1. The “good” action a_g deterministically leads to the next state in the chain, while the “bad” action deterministically leads to a terminal state. The only place where a non-zero reward can be received is the last state H , if the good action is chosen. The starting state is 1, and so **the only way** to receive non-zero reward is to select a_g for **all the H time steps** within the episode. (Since the length of the episode is also H , selecting actions uniformly brings **no information** about the optimal sequence of actions, unless by chance all of the actions sampled happen to be good; the probability that this occurs is exponentially small in H .) This means that T needs to be at least $O(2^H)$ to achieve nontrivial regret, and highlights the need for more strategic exploration.

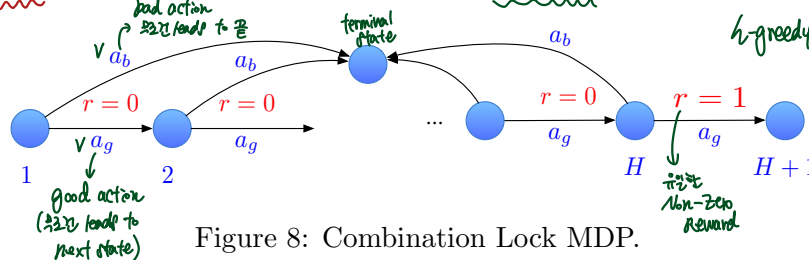


Figure 8: Combination Lock MDP.

Given the failure of ϵ -Greedy for this example, one can ask whether other algorithmic principles also fail. As we will show now, **the principle of optimism** succeeds, and an analogue of the UCB method yields a **regret bound** that is **polynomial** in the parameters $|\mathcal{S}|$, $|\mathcal{A}|$, and H . Before diving into the details, we present a collection of standard tools for analysis in MDPs, which will find use throughout the remainder of the lecture notes.

5.4 Analysis Tools

One of the most basic tools employed in the analysis of reinforcement learning algorithms is the **performance difference lemma**, which expresses the **difference in values** for **two policies** (in terms of **differences in single-step decisions** made by the two policies). The simple proof, stated below, proceeds by successively changing **one policy** (into **another**) and **keep track** of the **ensuing differences** (in expected rewards). One may also interpret this lemma as a version of the **credit assignment mechanism**. (정제-결과를 망친 "step"이 무엇냐~)

Henceforth, we adopt the following simplified notation. When a policy π is applied to the random variable s_h , we drop the subscript h and write $\pi(s_h)$ instead of $\pi_h(s_h)$, whenever this does not cause confusion.

Two Policies under same MDP

Lemma 13 (Performance Difference Lemma): For any $s \in \mathcal{S}$, and $\pi, \pi' \in \Pi_{\text{rs}}$,

$$\underbrace{V_1^{M, \pi'}(s) - V_1^{M, \pi}(s)}_{\text{difference in values for two } (\pi \text{ and } \pi') \text{ policies}} = \sum_{h=1}^H \mathbb{E}^{M, \pi} \left[\underbrace{Q_h^{M, \pi'}(s_h, \pi'(s_h)) - Q_h^{M, \pi'}(s_h, \underbrace{a_h}_{\pi(s_h)})}_{\text{differences in single-step decisions}} \mid s_1 = s \right] \quad (5.13)$$

Proof of Lemma 13. Fix a pair of policies π, π' and define

$$\pi^h = (\pi_1, \dots, \pi_{h-1}, \pi'_h, \dots, \pi'_H),$$

with $\pi^1 = \pi'$ and $\pi^H = \pi$. By telescoping, we can write

$$\begin{aligned} \text{ep1: } \pi^1 &= (\pi'_1, \pi'_2, \dots, \pi'_H) = \pi' & h=1 \text{ 일 때:} \\ \text{ep2: } \pi^2 &= (\pi_1, \pi'_2, \dots, \pi'_H) & \\ \text{ep3: } \pi^3 &= (\pi_1, \pi_2, \pi'_3, \dots, \pi'_H) & \\ & \vdots \pi \text{로 바꾸어 } \pi \text{로 바꿔나감 \& keep track of the 그미따는 result} & \\ \pi^H &= (\pi_1, \dots, \pi_H) = \pi & \end{aligned} \quad V_1^{M, \pi'}(s) - V_1^{M, \pi}(s) = \sum_{h=1}^H \left(V_1^{M, \pi^h}(s) - V_1^{M, \pi^{h+1}}(s) \right) \quad (5.14)$$

Observe that for each h , we have

$$\underbrace{V_1^{M, \pi^h}(s) - V_1^{M, \pi^{h+1}}(s)}_{\text{by def. of } V} = \left(\mathbb{E}^{M, \pi^h} \left[\sum_{t=1}^H r_t \mid s_1 = s \right] - \mathbb{E}^{M, \pi^{h+1}} \left[\sum_{t=1}^H r_t \mid s_1 = s \right] \right) \quad (5.15)$$

Here, one process evolves according to (M, π^h) and the one evolves according to (M, π^{h+1}) . The processes only differ in the action taken **once the state s_h is reached**. In the former, the action $\pi'(s_h)$ is taken, whereas in the latter it is $\pi(s_h)$. Hence, (5.15) is equal to

$$\mathbb{E}_{s_h \sim (M, \pi)} \mathbb{E}^{M, \pi} \left[\underbrace{Q_h^{M, \pi'}(s_h, \pi'(s_h)) - Q_h^{M, \pi'}(s_h, \underbrace{\pi(s_h)}_{\text{action value function의 form}})}_{\text{"SA" state에 갔을 때만 } \gamma \text{ 에 차가 생긴!}} \mid s_1 = s \right] \quad (5.16)$$

which can be written as

$$\mathbb{E}_{(s_h, a_h) \sim (M, \pi)} \mathbb{E}^{M, \pi} \left[Q_h^{M, \pi'}(s_h, \pi'(s_h)) - Q_h^{M, \pi'}(s_h, a_h) \mid s_1 = s \right]. \quad (5.17)$$

□

In contrast to the performance difference lemma, which relates the values of two policies under the same MDP, the **next result** relates the performance of the **same policy under two different MDPs**. Specifically, the difference in initial value for two MDPs is decomposed into a sum of errors between layer-wise value functions.

Performance of "Same policy" under two different MDPs. (첫번째파트만)

Lemma 14 (Bellman residual decomposition): For any pair of MDPs $M = (P^M, R^M)$ and $\widehat{M} = (P^{\widehat{M}}, R^{\widehat{M}})$, for any $s \in \mathcal{S}$, and policies $\pi \in \Pi_{\text{RNS}}$,

State Value function

$$V_1^{M,\pi}(s) - V_1^{\widehat{M},\pi}(s) = \sum_{h=1}^H \mathbb{E}^{\widehat{M},\pi} [Q_h^{M,\pi}(s_h, a_h) - r_h - V_{h+1}^{M,\pi}(s_{h+1}) \mid s_1 = s] \quad (5.18)$$

error

Hence, for $M, \widehat{M}$ with the same initial state distribution,

Value for policy π

$$f^M(\pi) - f^{\widehat{M}}(\pi) = \sum_{h=1}^H \mathbb{E}^{\widehat{M},\pi} [Q_h^{M,\pi}(s_h, a_h) - r_h - V_{h+1}^{M,\pi}(s_{h+1})]. \quad (5.19)$$

In addition, for any MDP M and function $Q = (Q_1, \dots, Q_H, Q_{H+1})$ with $Q_{H+1} \equiv 0$, letting $\pi_{Q,h}(s) = \arg \max_{a \in \mathcal{A}} Q_h(s, a)$, we have

$$\max_{a \in \mathcal{A}} Q_1(s, a) - V_1^{M,\pi_Q}(s) = \sum_{h=1}^H \mathbb{E}^{M,\pi_Q} [Q_h(s_h, a_h) - [\mathcal{T}_h^M Q_{h+1}](s_h, a_h) \mid s_1 = s]. \quad (5.20)$$

and, hence,

$$\mathbb{E}_{s_1 \sim d_1} [\max_{a \in \mathcal{A}} Q_1(s_1, a)] - f^M(\pi_Q) = \sum_{h=1}^H \mathbb{E}^{M,\pi_Q} [Q_h(s_h, a_h) - [\mathcal{T}_h^M Q_{h+1}](s_h, a_h)]. \quad (5.21)$$

Note that for the second part of **Lemma 14** $Q = (Q_1, \dots, Q_H)$ can be any sequence of functions, and need not be a value function corresponding to a particular policy or MDP. (It is worth noting that Q gives rise to the greedy policy π_Q , which, in turn, gives rise to Q^{M,π_Q} (the value of π_Q in model M), but it may well be the case that $Q^{M,\pi_Q} \neq Q$.)

Proof of Lemma 14. We will prove (5.19), and omit the proof for (5.18), which is similar but more verbose. We have

(5.19의 proof)
(RHS를 먼저만들어놓고
그걸 LHS 쪽으로 변형해감)

$$\begin{aligned} \sum_{h=1}^H \mathbb{E}^{\widehat{M},\pi} [Q_h^{M,\pi}(s_h, a_h) - \underbrace{r_h}_{\text{이게 빼요}} - V_{h+1}^{M,\pi}(s_{h+1})] &= \sum_{h=1}^H \mathbb{E}^{\widehat{M},\pi} [Q_h^{M,\pi}(s_h, a_h) - V_{h+1}^{M,\pi}(s_{h+1})] - \mathbb{E}^{\widehat{M},\pi} \left[\sum_{h=1}^H r_h \right] \\ &= \sum_{h=1}^H \mathbb{E}^{\widehat{M},\pi} [Q_h^{M,\pi}(s_h, a_h) - V_{h+1}^{M,\pi}(s_{h+1})] - \underbrace{f^{\widehat{M}}(\pi)}_{f \text{의 정의에 의해}}. \\ &\quad \downarrow V_h^{M,\pi}(s) = \mathbb{E}_{a \sim \pi_h(s)} [Q_h^{M,\pi}(s, a)] \\ &= \sum_{k=1}^H \mathbb{E}^{\widehat{M},\pi} [V_k^{M,\pi}(s_k) - V_{k+1}^{M,\pi}(s_{k+1})] \\ &= \mathbb{E}^{\widehat{M},\pi} [V_1^{M,\pi}(s_1)] - \mathbb{E}^{\widehat{M},\pi} [V_{H+1}^{M,\pi}(s_{H+1})] \\ &= f^M(\pi) \end{aligned}$$

On the other hand, since $V_h^{M,\pi}(s) = \mathbb{E}_{a \sim \pi_h(s)}[Q_h^{M,\pi}(s, a)]$, a telescoping argument yields

$$\begin{aligned} \sum_{h=1}^H \mathbb{E}^{\widehat{M}, \pi} [Q_h^{M,\pi}(s_h, a_h) - V_{h+1}^{M,\pi}(s_{h+1})] &= \sum_{h=1}^H \mathbb{E}^{\widehat{M}, \pi} [V_h^{M,\pi}(s_h) - V_{h+1}^{M,\pi}(s_{h+1})] \\ &= \mathbb{E}^{\widehat{M}, \pi} [V_1^{M,\pi}(s_1)] - \mathbb{E}^{\widehat{M}, \pi} [V_{H+1}^{M,\pi}(s_{H+1})] \\ &= f^M(\pi), \end{aligned}$$

$* V_{H+1}^{M,\pi} = 0.$

where we have used that $V_{H+1}^{M,\pi} = 0$, and that both MDPs have the same initial state distribution. We prove (5.21) (omitting the proof of (5.20)) using a similar argument. We have

$$\begin{aligned} &\sum_{h=1}^H \mathbb{E}^{M, \pi_Q} [Q_h(s_h, a_h) - \underbrace{r_h}_{\text{State, Action}} - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, a)] \\ &\stackrel{\text{RHS}}{=} \sum_{h=1}^H \mathbb{E}^{M, \pi_Q} [Q_h(s_h, a_h) - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, a)] - \mathbb{E}^{M, \pi_Q} \left[\sum_{h=1}^H r_h \right] \\ &= \sum_{h=1}^H \mathbb{E}^{M, \pi_Q} [Q_h(s_h, a_h) - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, a)] - f^M(\pi_Q). \end{aligned}$$

by 정의

Since $a_{h+1} = \pi_{Q,h}(s_{h+1}) = \arg \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, a)$, we have

$$\mathbb{E}^{M, \pi_Q} [Q_h(s_h, a_h) - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, a)] = \mathbb{E}^{M, \pi_Q} [Q_h(s_h, a_h) - Q_{h+1}(s_{h+1}, a_{h+1})],$$

and the result follows by telescoping. $\sum_{h=1}^H \mathbb{E}^{M, \pi_Q} [Q_h(s_h, a_h) - Q_{h+1}(s_{h+1}, a_{h+1})] = \mathbb{E}^{M, \pi_Q} [Q_1(s_1, a_1)] - \mathbb{E}^{M, \pi_Q} [Q_{H+1}(s_{H+1}, a_{H+1})]$
 $= \mathbb{E}_{s_1, a_1} [\max_{a \in \mathcal{A}} Q_1(s_1, a)] - 0$

Another similar analysis tool for MDPs, the *simulation lemma*, is deferred to Section 6 (Lemma 23). This result can be proven as a consequence of Lemma 14.

5.5 Optimism *optimism in the face of uncertainty (k2) : 1.7의 width가 shrink 하는 confidence interval을 만족하면, Regret이 bound 됨.*

To develop algorithms for **regret minimization** in **unknown** MDPs, we turn to the **principle of optimism**, which we have seen is successful in tackling multi-armed bandits and linear bandits (in small dimension). (Recall that for bandits, Lemma 7 gave a way to decompose the regret of optimistic algorithms into width of confidence intervals.) What is the **analogue of Lemma 7** for MDPs? Thinking of optimistic estimates at the level of expected rewards for policies π is unwieldy, and we need to dig into the structure of these **multi-stage decisions**. In particular, the approach we employ is to construct a **sequence of optimistic value functions** $\bar{Q}_1, \dots, \bar{Q}_H$ which are **guaranteed to over-estimate** the **optimal value function** $Q^{M,*}$. (For multi-armed bandits, **implementing optimism** amounted to adding “bonuses,” constructed from past data, to estimates (for the reward function).) We will construct optimistic value functions in a similar fashion. Before giving the construction, we introduce a technical lemma, which **quantifies the error** (in using such optimistic estimates) (in terms of **Bellman residuals**): Bellman residuals **measure self-consistency** (of the optimistic estimates) under the application of the Bellman operator.

optimistic estimates를 사용하는 데 있어
 생기는 error를 양분화하는!
 in terms of “Bellman Residuals”
 $V^* - \bar{V}_h$ $\hookrightarrow$ Measure “Self Consistency” of
 optimistic estimates
 (under the T of Bellman operator)

Lemma 14
(In terms of Bellman Residuals)

Lemma 15 (Error decomposition for optimistic policies): Let $\{\bar{Q}_1, \dots, \bar{Q}_H\}$ be a sequence of functions $\bar{Q}_h: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ with the property that for all (s, a) ,
(Sequence of OPTIMISTIC Value functions)

①

$$Q_h^{M,*}(s, a) \leq \bar{Q}_h(s, a) \quad \text{Optimistic} \quad (5.22)$$

and set $\bar{Q}_{H+1} \equiv 0$. Let $\hat{\pi} = (\hat{\pi}_1, \dots, \hat{\pi}_H)$ be such that $\hat{\pi}_h(s) = \arg \max_a Q_h(s, a)$. Then for all $s \in \mathcal{S}$,
Bellman Residual이 "자기가" $\bar{Q}_h$ 가 "self consistent"

②

$$V_1^{M,*}(s) - V_1^{M,\hat{\pi}}(s) \leq \sum_{h=1}^H \mathbb{E}^{M,\hat{\pi}}[(\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1})(s_h, \hat{\pi}(s_h)) \mid s_1 = s]. \quad (5.23)$$

RHS $\downarrow$ implies LHS $\downarrow$

Lemma 15 tells us that closeness of $\bar{Q}_h$ to the Bellman backup $\mathcal{T}_h^M \bar{Q}_{h+1}$ implies closeness of $\hat{\pi}$ to π_M (in terms of the value). As a sanity check, if $\bar{Q}_h = Q_h^{M,*}$, the right-hand side of (5.23) is zero, since $Q_h^{M,*} = \mathcal{T}_h^M Q_{h+1}^{M,*}$. Crucially, errors do not accumulate too fast as a function of the horizon. This fact should not be taken for granted: in general, if $\bar{Q}$ is not optimistic, it could have been the case that small changes in $\bar{Q}_h$ exponentially degrade the quality of the policy $\hat{\pi}$.

Another important aspect of the decomposition (5.23) is the *on-policy* nature of the terms in the sum. Observe that the law of s_h for each of the terms is given by executing $\hat{\pi}$ in model M . The distribution of s_h is often referred to as the *roll-in distribution*; when this distribution is induced by the policy executed by the algorithm we may have a better control of the error than in the *off-policy* case when the roll-in distribution is given by π_M or another unknown policy.

Proof of Lemma 15. Let $\bar{V}_h(s) := \max_{a \in \mathcal{A}} \bar{Q}_h(s, a)$. Just as in the proof of Lemma 7, the assumption that $\bar{Q}_h$ is "optimistic" implies that

$$Q_h^{M,*}(s_h, \pi_M(s_h)) \leq \bar{Q}_h(s_h, \pi_M(s_h)) \leq \bar{Q}_h(s_h, \hat{\pi}(s_h))$$

and, hence, $V_1^{M,*}(s) \leq \bar{V}_1(s)$. Then, (5.20) applied to $Q = \bar{Q}$ and $\pi_Q = \hat{\pi}$ states that

$$\bar{V}_1(s) - V_1^{M,\hat{\pi}}(s) = \sum_{h=1}^H \mathbb{E}^{M,\hat{\pi}}[\bar{Q}_h(s_h, a_h) - [\mathcal{T}_h^M \bar{Q}_{h+1}](s_h, a_h) \mid s_1 = s]. \quad (5.24)$$

$$(5.20) \quad \max_{a \in \mathcal{A}} Q_1(s, a) - V_1^{M,\pi_Q}(s) = \sum_{h=1}^H \mathbb{E}^{M,\pi_Q}[Q_h(s_h, a_h) - [\mathcal{T}_h^M Q_{h+1}](s_h, a_h) \mid s_1 = s]. \quad (5.20)$$

□

Remark 16: In fact, the proof of Lemma 15 only uses that the initial value $\bar{Q}_1$ is optimistic. However, to construct a value function with this property, the algorithms we consider will proceed by backwards induction, producing optimistic estimates $\bar{Q}_1, \dots, \bar{Q}_H$ in the process.

5.6 The UCB-VI Algorithm for Tabular MDPs

Principle of Optimism을 적용
Tabular MDP 이니셜 $\leftarrow$ State & Action spaces가 작음.
online RL 이니셜
regret bound를 계산! $\leftarrow \mathcal{S}, \mathcal{A}, H$ 이 finite polynomial!

We now instantiate the principle of optimism to give regret bounds for online reinforcement learning in *tabular MDPs*. Tabular RL may be thought of as an analogue of finite-armed

bandits: we assume no structure across states and actions, but require that the state and action spaces are small. The regret bounds we present will depend polynomially (on $S = |\mathcal{S}|$ and $A = |\mathcal{A}|$, as well as the horizon H .)

Preliminaries. For simplicity, we assume that the reward function is known to the learner, so that only the transition probabilities are unknown. This does not change the difficulty of the problem in a meaningful way, but allows us to keep notation light.

Reward func: 알려지 있음.
모르는 것: Transition 확률!

(for simplicity, reward is known & $V \in [0, 1]$)

Assumption 6: Rewards are deterministic, bounded, and known to the learner: $R_h^M(s, a) = \delta_{r_h(s, a)}$ for known $r_h : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, for all M . In addition, assume for simplicity that $V_1^{M, \star}(s) \in [0, 1]$ for any $s \in \mathcal{S}$.

Define, with a slight abuse of notation,

n : frequency

$$n_h^t(s, a) = \sum_{i=1}^{t-1} \mathbb{I}\{(s_h^i, a_h^i) = (s, a)\}, \quad \text{and} \quad n_h^t(s, a, s') = \sum_{i=1}^{t-1} \mathbb{I}\{(s_h^i, a_h^i, s_{h+1}^i) = (s, a, s')\},$$

state action state action next state

각 episode 별 ... 빈도 수 셈.
($t-1$)

as the empirical state-action and state-action-next state frequencies. We can estimate the transition probabilities via

transition 확률에 대한
측정! (빈도)

(empirical!)

$$\hat{P}_h^t(s' | s, a) = \frac{n_h^t(s, a, s')}{n_h^t(s, a)}. \quad (5.25)$$

The UCB-VI algorithm. The following algorithm, UCB-VI ("Upper Confidence Bound Value Iteration") [16], combines the notion of optimism with dynamic programming.

UCB-VI Upper Confidence Bound Value Iteration

for $t = 1, \dots, T$ do t : episode

Let $\bar{V}_{H+1}^t \equiv 1$. terminal (Horizon)

for $h = H, \dots, 1$ do horizon (step..)

Update $n_h^t(s, a)$, $n_h^t(s, a, s')$, and $b_{h, \delta}^t(s, a)$, for all $(s, a) \in \mathcal{S} \times \mathcal{A}$.

// $b_{h, \delta}^t(s, a)$ is a bonus computed in (5.27).

Compute:

(state-)action value func.

$$\bar{Q}_h^t(s, a) = \left\{ r_h(s, a) + \mathbb{E}_{s' \sim \hat{P}_h^t(\cdot | s, a)} \bar{V}_{h+1}^t(s') + b_{h, \delta}^t(s, a) \right\} \wedge 1. \quad (5.26)$$

state value func. immediate reward future reward

Set $\bar{V}_h^t(s) = \max_{a \in \mathcal{A}} \bar{Q}_h^t(s, a)$ and $\hat{\pi}_h^t(s) = \arg \max_{a \in \mathcal{A}} \bar{Q}_h^t(s, a)$.

Collect trajectory $(s_1^t, a_1^t, r_1^t), \dots, (s_H^t, a_H^t, r_H^t)$ according to $\hat{\pi}^t$.

of that episode (trajectory)

The UCB-VI algorithm will be analyzed using Lemma 15. In constructing functions $\bar{Q}_h$, we will need to satisfy two goals: (1) ensure that with high probability (5.22) is satisfied, i.e.

①

$\bar{Q}_h$ s are optimistic; and (2) that $\bar{Q}_h$ s are "self-consistent", in the sense that the Bellman residuals in (5.23) are small. The second requirement already suggests that we should define

②

$\bar{Q}_h$ approximately as a Bellman backup $\mathcal{T}_h^M \bar{Q}_{h+1}$ (going backwards for $h = H + 1, \dots, 1$ as in dynamic programming, while ensuring the first requirement.) In addition to these considerations, we will have to use a surrogate for the "Bellman operator" $\mathcal{T}_h^M$, since the model

대역 $\hookrightarrow \mathcal{T}_h^M$

Reward func: 알려져 있음.
 M is not known. This is achieved by estimating M using empirical transition frequencies. Putting these ideas together gives the update in (5.26). We apply the principle of value iteration, except that

1. For each episode t , we augment the rewards $r_h(s, a)$ with a "bonus" $b_{h,\delta}^t(s, a)$ (designed to ensure optimism).

2. The Bellman operator is approximated using the estimated transition probabilities in (5.25).

The bonus functions play precisely the same role as the width of the confidence interval in (2.19): these bonuses ensure that (5.22) holds with high probability, as we will show below in Lemma 16.

The following theorem shows that with an appropriate choice of bonus, this algorithm achieves a polynomial regret bound.

Theorem 1: For any $\delta > 0$, UCB-VI with

$$b_{h,\delta}^t(s, a) = 2\sqrt{\frac{\log(2SAHT/\delta)}{n_h^t(s, a)}} \quad (5.27)$$

guarantees that with probability at least $1 - \delta$,

$$\text{Regret} \lesssim HS\sqrt{AT} \cdot \sqrt{\log(SAHT/\delta)}$$

We mention that a slight variation on Lemma 18 below (using the Freedman inequality instead of the Azuma-Hoeffding inequality) yields an improved rate of $O(H\sqrt{SAT} + \text{poly}(H, S, A) \log T)$, and the optimal rate can be shown to be $\Theta(\sqrt{HSAT})$; this is achieved through a more careful choice for the bonus $b_{h,\delta}^t$ and a more refined analysis. We remark that care should be taken in comparing results in the literature, as scaling conventions for the individual and cumulative rewards (as in Assumption 6) can vary.

5.6.1 Analysis for a Single Episode (within one episode..t) (FIX t)

Our aim is to bound the regret

$$\text{Reg} = \sum_{t=1}^T f^M(\pi_{M^*}) - f^M(\pi^t)$$

for UCB-VI. To do so, we first prove several helper lemmas concerning the performance within each episode t . In what follows, we fix t and drop the superscript t .

Given the estimated transitions $\{\hat{P}_h(\cdot | s, a)\}_{h,s,a}$, define the estimated MDP $\hat{M} = \{S, \mathcal{A}, \{\hat{P}_h\}_{h=1}^H, \{R_h^M\}_{h=1}^H, d_1\}$. The associated "Bellman operator" is defined as..

$$\hat{T}_h^{\hat{M}} Q(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim \hat{P}_h(\cdot | s, a)} \max_a Q(s', a) \quad (5.28)$$

for $Q : S \times \mathcal{A} \rightarrow \mathbb{R}$. Consider the sequence of functions $\bar{Q}_h : S \times \mathcal{A} \rightarrow [0, 1]$, $\bar{V}_h : S \rightarrow [0, 1]$, for $h = 1, \dots, H+1$, (with $\bar{Q}_{H+1} \equiv 0$) and

$$\bar{Q}_h(s, a) = \left\{ \bar{T}_h^{\hat{M}} \bar{Q}_{h+1}(s, a) + b_{h,\delta}(s, a) \right\} \wedge 1, \quad \text{and} \quad \bar{V}_h(s) = \max_a \bar{Q}_h(s, a) \quad (5.29)$$

for bonus functions $b_{h,\delta} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ to be chosen later. Henceforth, we follow the usual notation that for functions f, g over the same domain, $f \leq g$ indicates pointwise inequality over the domain.

The first lemma we present shows that as long as the bonuses $b_{h,\delta}$ are large enough to bound the error (between the estimated transition probabilities and true transition probabilities,) the functions $\bar{Q}_1, \dots, \bar{Q}_H$ constructed above will be optimistic.

Lemma 16: Suppose we have estimates $\{\hat{P}_h(\cdot | s, a)\}_{h,s,a}$ and a function $b_{h,\delta} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ with the property that for all $s \in \mathcal{S}, a \in \mathcal{A}$,

$$\left| \sum_{s'} \hat{P}_h^V(s' | s, a) V_h^{M,*}(s') - \sum_{s'} P_h^M(s' | s, a) V_h^{M,*}(s') \right| \leq b_{h,\delta}(s, a). \quad (5.30)$$

Then for all $h \in [H]$, we have

$$\bar{Q}_h \geq Q_h^{M,*}, \quad \text{and} \quad \bar{V}_h \geq V_h^{M,*} \quad (5.31)$$

for $\bar{Q}_h, \bar{V}_h$ defined in (5.29).

Proof of Lemma 16. The proof proceeds by backward induction on the statement

$$\bar{V}_h \geq V_h^{M,*}$$

with $h = H + 1$ down to $h = 1$. We start with the base case $h = H + 1$, which is trivial because $\bar{V}_{H+1} = V_{H+1}^{M,*} \equiv 0$. Now, assume $\bar{V}_{h+1} \geq V_{h+1}^{M,*}$, and let us prove the induction step. Fix $(s, a) \in \mathcal{S} \times \mathcal{A}$. If $\bar{Q}_h(s, a) = 1$, then, trivially, $\bar{Q}_h(s, a) \geq Q_h^{M,*}(s, a)$. Otherwise, $\bar{Q}_h(s, a) = \mathcal{T}_h^M \bar{Q}_{h+1}(s, a) + b_{h,\delta}(s, a)$, and thus

$$\begin{aligned} \bar{Q}_h(s, a) - Q_h^{M,*}(s, a) &= b_{h,\delta}(s, a) + \mathbb{E}_{s' \sim \hat{P}_h(\cdot | s, a)} \bar{V}_{h+1}(s') - \mathbb{E}_{s' \sim P_h^M(\cdot | s, a)} V_{h+1}^{M,*}(s') \\ &\geq b_{h,\delta}(s, a) + \mathbb{E}_{s' \sim \hat{P}_h(\cdot | s, a)} \bar{V}_{h+1}^{M,*}(s') - \mathbb{E}_{s' \sim P_h^M(\cdot | s, a)} V_{h+1}^{M,*}(s') \geq 0. \end{aligned}$$

This, in turn, implies that $\bar{V}_h(s) = \max_a \bar{Q}_h(s, a) \geq \max_a Q_h^{M,*}(s, a) = V_h^{M,*}(s)$, concluding the induction step. $\square$

We now analyze the effect of using an estimated model $\hat{M}$ for the Bellman operator (rather than the true unknown $\mathcal{T}_h^M$).

Lemma 17: Suppose we have estimates $\{\hat{P}_h(\cdot | s, a)\}_{h,s,a}$ and $b'_{h,\delta}(s, a) : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ with the property that

$$\max_{V \in \{0,1\}^{\mathcal{S}}} \left| \sum_{s'} \hat{P}_h^V(s' | s, a) V(s') - \sum_{s'} P_h^M(s' | s, a) V(s') \right| \leq b'_{h,\delta}(s, a) \quad (5.32)$$

Then the Bellman residual satisfies

$$\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1} \leq (b_{h,\delta} + b'_{h,\delta}) \wedge 1. \quad (5.33)$$

for $\bar{Q}_h, \bar{V}_h$ defined in (5.29).

Proof of Lemma 17. That $\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1} \leq 1$ is immediate. To prove the main result, observe that

$$\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1} \stackrel{\text{def.}}{=} \left\{ \mathcal{T}_h^M \bar{Q}_{h+1} + b_{h,\delta} \right\} \wedge 1 - \mathcal{T}_h^M \bar{Q}_{h+1} \stackrel{1..}{\leq} (\mathcal{T}_h^M - \mathcal{T}_h^M) \bar{Q}_{h+1} + b_{h,\delta} \stackrel{\infty}{\leq} b'_{h,\delta} \quad (5.34)$$

For any $Q \in \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$,

$$(\mathcal{T}_h^M - \mathcal{T}_h^M) Q(s, a) \stackrel{\text{def.}}{=} \mathbb{E}_{s' \sim \hat{P}_h(\cdot|s,a)} \max_a Q(s', a) - \mathbb{E}_{s' \sim P_h^M(\cdot|s,a)} \max_a Q(s', a) \quad (5.35)$$

$$\leq \max_{V \in [0,1]^S} |\mathbb{E}_{s' \sim \hat{P}_h(\cdot|s,a)} V(s') - \mathbb{E}_{s' \sim P_h^M(\cdot|s,a)} V(s')| \stackrel{(5.32) \text{ of given property}}{\leq} b'_{h,\delta}(s,a) \quad (5.36)$$

Since the maximum is achieved at a vertex of $[0, 1]^S$, the statement follows. $\square$

5.6.2 Regret Analysis

We now bring back the time index t and show that the estimated transition probabilities (in UCB-VI) satisfy conditions (of Lemma 16 and Lemma 17), ensuring that the functions $\bar{Q}_1^t, \dots, \bar{Q}_H^t$ are optimistic.

Lemma 18: Let $\{\hat{P}_h^t\}_{h \in [H], t \in [T]}$ be defined as in (5.25). Then with probability at least $1 - \delta$, the functions

$$b_{h,\delta}^t(s, a) = 2 \sqrt{\frac{\log(2SAHT/\delta)}{n_h^t(s, a)}}, \quad \text{and} \quad b'_{h,\delta}(s, a) = 8 \sqrt{\frac{S \log(2SAHT/\delta)}{n_h^t(s, a)}}$$

satisfy the assumptions of Lemma 16 and Lemma 17, respectively, for all $s \in \mathcal{S}$, $a \in \mathcal{A}$, $h \in [H]$, and $t \in [T]$ simultaneously. $\square$

Proof of Lemma 18. We leave the proof as an exercise. $\square$

Proof of Theorem 1. Putting everything together, we can now prove Theorem 1. Under the event in Lemma 18, the functions $\bar{Q}_1^t, \dots, \bar{Q}_H^t$ are optimistic, which means that the conditions of Lemma 15 hold, and the instantaneous regret on round t (conditionally on $s_1 \sim d_1$) is at most

$$\stackrel{\text{RHS of Lemma 15}}{\leq} \sum_{h=1}^H \mathbb{E}^{M, \hat{\pi}^t} [(\bar{Q}_h^t - \mathcal{T}_h^M \bar{Q}_{h+1}^t)(s_h^t, \hat{\pi}_h^t(s_h^t)) \mid s_1 = s] \stackrel{\text{first inequality: Lemma 15}}{\leq} \sum_{h=1}^H \mathbb{E}^{M, \hat{\pi}^t} [(b_{h,\delta}(s_h^t, \hat{\pi}_h^t(s_h^t)) + b'_{h,\delta}(s_h^t, \hat{\pi}_h^t(s_h^t))) \wedge 1],$$

where the second inequality invokes Lemma 17. Summing over $t = 1, \dots, T$, and applying the Azuma-Hoeffding inequality, we have that with probability at least $1 - \delta$, the regret of UCB-VI is bounded by

$$\begin{aligned} & \sum_{t=1, \dots, T} \left(\sum_{h=1}^H \mathbb{E}^{M, \hat{\pi}^t} [(b_{h,\delta}(s_h^t, \hat{\pi}_h^t(s_h^t)) + b'_{h,\delta}(s_h^t, \hat{\pi}_h^t(s_h^t))) \wedge 1] \right) \\ & \lesssim \sum_{t=1}^T \sum_{h=1}^H (b_{h,\delta}(s_h^t, \hat{\pi}_h^t(s_h^t)) + b'_{h,\delta}(s_h^t, \hat{\pi}_h^t(s_h^t))) \wedge 1 + \sqrt{HT \log(1/\delta)}. \end{aligned}$$

Azuma-Hoeffding

appropriate choice of bonus γ

$$b_{h,\delta}^t(s,a) = 2\sqrt{\frac{\log(2SAHT/\delta)}{n_h^t(s,a)}} \quad \text{!}$$

이렇게 정해줌.

Using the **bonus** definition in (5.27), the **bonus** term above is bounded by $\frac{1}{\sqrt{n_h^t(s,a)}}$

$$\sum_{t=1}^T \sum_{h=1}^H \sqrt{\frac{S \log(2SAHT/\delta)}{n_h^t(s_h^t, \hat{\pi}_h^t(s_h^t))}} \wedge 1 \leq \sqrt{S \log(2SAHT/\delta)} \sum_{t=1}^T \sum_{h=1}^H \frac{1}{\sqrt{n_h^t(s_h^t, \hat{\pi}_h^t(s_h^t))}} \wedge 1 \quad (5.37)$$

이렇게 풀고

The double summation can be handled in the same fashion as Lemma 8:

$$\sum_{t=1}^T \sum_{h=1}^H \frac{1}{\sqrt{n_h^t(s_h^t, \hat{\pi}_h^t(s_h^t))}} \wedge 1 = \sum_{h=1}^H \left(\sum_{(s,a)} \sum_{t=1}^T \frac{\mathbb{I}\{(s_h^t, \hat{\pi}_h^t(s_h^t)) = (s,a)\}}{\sqrt{n_h^t(s,a)}} \right) \wedge 1$$

*Lemma 8 proof...

$$\sum_{t=1}^T \frac{1}{\sqrt{t+1}} \wedge 1 \leq 1 + 2\sqrt{\pi}$$

$$\lesssim \sum_{h=1}^H \sum_{(s,a)} \sqrt{n_h^T(s,a)} \leq H\sqrt{SAT}.$$

→ Regret is Bounded

6. GENERAL DECISION MAKING

So far, we have covered three general frameworks for interaction decision making: The contextual bandit problem, the structured bandit problem, and the episodic reinforcement learning problem; all of these frameworks generalize the classical multi-armed bandit problem in different directions. In the context of structured bandits, we introduced a complexity measure called the Decision-Estimation Coefficient (DEC), which gave a generic approach to algorithm design, and allowed us to reduce the problem of interactive decision making to that of supervised online estimation. In this section, we will build on this development on two fronts: First, we will introduce a unified framework for decision making, which subsumes all of the frameworks we have covered so far. Then, we will show that i) the Decision-Estimation Coefficient and its associated meta-algorithm (E2D) extend to the general decision making framework, and ii) boundedness of the DEC is not just sufficient, but actually *necessary* for low regret, and thus constitutes a fundamental limit. As an application of the general tools we introduce, we will show how to use the (generalized) Decision-Estimation Coefficient to solve the problem of tabular reinforcement learning (Section 6.6), offering an alternative to the UCB-VI method we introduced in Section 5.

6.1 Setting

For the remainder of the course, we will focus on a framework called Decision Making with Structured Observations (DMSO), which subsumes all of the decision making frameworks we have encountered so far. The protocol proceeds in T rounds, where for each round $t = 1, \dots, T$:

1. The learner selects a *decision* $\pi^t \in \Pi$, where Π is the *decision space*.
2. Nature selects a *reward* $r^t \in \mathcal{R}$ and *observation* $o^t \in \mathcal{O}$ based on the decision, where $\mathcal{R} \subseteq \mathbb{R}$ is the *reward space* and $\mathcal{O}$ is the *observation space*. The reward and observation are then observed by the learner.

We focus on a stochastic variant of the DMSO framework.